

CV Analysis & Viva Map — Line by Line

CASSA RA 2026-27 · "every line on your CV is fair game"

The viva is "about what you have submitted." This walks through your CV section by section: for each entry, the **question they may ask**, a **model answer**, and the **landmines** to prepare for. Read it once out loud.

1. Header / Research Interests

Listed: Supernova Cosmology, Host-Galaxy SED Fitting, Bayesian Inference, Radio AGN Classification, ML in Astrophysics, Time-Domain Astronomy. - **Q: "Which of these is your main interest?"** → *Supernova cosmology and host-galaxy systematics; the others feed into it (Bayesian inference is the method, ML and time-domain are the tools and the data).* - **Landmine:** do not list an interest you cannot discuss. You can defend every one of these; keep the answer tied to your SN work.

2. Education — RUET, BSc ECE (Feb 2019 to Sept 2024), CGPA 2.88/4.00

- **Q (almost certain): "Your CGPA is modest. Why?"** → *I put my energy into research and olympiads rather than coursework, and I taught myself astrophysics with no astro faculty at RUET. My output, an IEEE publication, an A&A paper, and three first-author manuscripts, shows where I focused. My research trajectory is steeply upward. (Honest, forward-looking, no excuses.)*
- **Q: "Why ECE, not Physics?"** → *ECE gave me signal processing, numerical methods, and machine learning, which is exactly the toolkit modern observational cosmology runs on. I added the astrophysics myself.*
- **IELTS 7.0 (S: 6.0):** unlikely to be probed; if asked, it clears every requirement.
- **Landmine, the gap:** thesis ended Sept 2024, CASSA internship began Oct 2025. Have a clean answer: *I finalised and published PIE-ADA, sat IELTS (Sept 2025), started the LunaXNet collaboration, and self-studied SED fitting before joining CASSA.*

3. Research Experience (the heart of the viva)

3a. Postbacc RA, CASSA (Mar-Jun 2026), Supervisor Dr. Uddin

CV claims: lead analyst, replaced emcee with Nautilus, 5-sampler benchmark, CSP-I 7-band, age step $+0.032 \pm 0.064$ mag, 3 first-author papers. - **Q: "You call yourself lead analyst, what exactly did you do?"** → *I own the analysis end to end: I rebuilt the Rose et al. 2019 mc-age pipeline, swapped in Nautilus on identical likelihood/priors/model, ran the sampler benchmark, extended it to CSP-I, and measured the age step.* - **Q: "Explain the age step number."** → *0.032 mag difference between old- and young-host SN, but the error is 0.064, so it is 0.50σ , consistent with zero, so no significant age bias, against Son et al. 2025. (Full physics in Note 01.)* - **Q: "Why Nautilus over emcee?"** → *NN-boosted importance nested sampling learns the posterior, $\sim 60\times$ fewer likelihood calls. (Math in Note 07.)* - **Landmine:** know your **priors and likelihood** and be honest about what is preliminary. Say "in preparation," do not oversell.

3b. Research Intern, CASSA (Oct 2025-Feb 2026), Supervisor Dr. Asad — RGC

CV: data curation + catalogue prep for RGC v1 (A&A accepted); guided v2 annotation tool. - **Q: "What was YOUR contribution to the A&A paper?"** → *Be precise and honest: data curation and catalogue preparation, cross-matching and source validation for the training set; not the network design. Do not claim the deep-learning model.* - **Q: "What is RGC classifying?"** → *radio-AGN morphology (FR I vs FR II) with a semi-supervised CNN. (FR I/II + synchrotron in Note 03; CNN in Note 07.)*

3c. Research Collaborator, RUET (2024-present) — LunaXNet

CV: hybrid CNN-Transformer for lunar mineral segmentation, Chandrayaan-1 M3 hyperspectral (85 bands, 430-3000 nm). - **Q: "Why a Transformer for hyperspectral data?"** → *The 85 spectral bands are a sequence; self-attention captures long-range relationships between bands (mineral absorption features) that a CNN alone misses.* (VAE/attention math in Note 07.)

3d. Undergraduate Thesis (Jan-Sept 2024) — PIE-ADA

CV: 14-class transient classification, PLAsTiCC (1.4M obs), physics-informed augmentation, 271 features, LightGBM log-loss 0.5763, 80.33% acc, 24-49% better, IEEE QPAIN 2026. - **Q: "Why physics-informed augmentation?"** → *Real surveys see redshifted, dust-reddened, noisy light curves. I augment with the actual physics (time dilation, extinction, correlated noise), so the model learns real invariances, not random jitter.* - **Q: "Why LightGBM and not a neural net?"** → *Tabular engineered features + class imbalance + limited data favour gradient-boosted trees; they were faster and more accurate here.* (Gradient boosting in Note 07.) - **Q: "What is weighted log-loss?"** → the PLAsTiCC metric that penalises confident wrong probabilities and reweights rare classes. (Formula in Note 07.)

4. Publications

- **Published:** PIE-ADA (IEEE QPAIN 2026) — you are first author, know it cold.
- **Accepted:** RGC (A&A 2026) — co-author; state your contribution honestly (Section 3b).
- **In prep (first author):** P1 sampler systematics (ApJ), P2 age step (ApJL), P3 pipeline+CANDLE (ApJS), LunaXNet (IEEE TGRS).
- **Q: "Which journal is which, and why?"** → P1 methods → ApJ; P2 short high-impact result → ApJL; P3 software → ApJS. Shows you understand the field's venues.
- **Landmine:** do not call an in-prep paper "published." Be exact.

5. Technical Skills

Python (5+ yr), C/C++, MATLAB, Bash, LaTeX; FSPS, Nautilus, MultiNest, Astropy, TOPCAT, DS9; PyTorch, TensorFlow, LightGBM, CNN, Transformer, Grad-CAM; knackbook web app. - **Q: "Pick a tool and explain it."** → Be ready for FSPS (SPS forward model), Nautilus (sampler), Astropy (coords/FITS), Grad-CAM (which pixels a CNN used). Full explanations in Notes 05 and 07. - **Landmine:** anything listed here is fair game; if you truly cannot discuss a tool, do not list it. You can defend all of these.

6. Awards, Leadership & Outreach

10BISS + I-HOW (Armenia 2026); IOAA 2017 national team; 12th BdAO National Winner; 8th BdPhO Regional Champion; University Physics Competition Silver (2020, 2023); UGC Scholarship; President, Astronomy & Science Society of RUET (200+ members). - **Q: "Tell me about IOAA / the olympiad path."** → your origin story: self-found competitions, junior mentors, national team; ties to the SOP. - **Q: "What did you do as society President?"** → led 200+ members, ran school science-outreach camps; shows leadership and communication, useful for an RA. - **Q: "What do you expect from I-HOW / Byurakan?"** → hands-on radio astronomy skills, first international selection since 2017; complements the RGC radio work.

The 8 questions you will almost certainly face (rehearse aloud)

1. Explain your supernova project in one minute. (Note 01)
2. Why does host-galaxy age matter for cosmology, and how do you test it?
3. Your CGPA is 2.88, why should we not worry?
4. What exactly was your contribution to the RGC A&A paper?
5. Why Nautilus over emcee, in plain terms? (Note 07)
6. What is your role vs your supervisor's in the SN project?
7. Where do you want to be in five years? → *PhD in supernova cosmology / computational astrophysics, building on this analysis.*

8. Why CASSA, and why should we pick you? → *I found this field without being shown the path; give me the environment and I will bring the host-age analysis to publication.*

Closing tips

- Bring printed **CV, SOP, and these notes**. Skim the CV the morning of.
- For every claim on the CV, have one sentence of "what I did" and one of "what it means."
- Be honest about scope (co-author vs lead, in-prep vs published). Panels respect precision, not inflation.
- If stuck: state what you know, reason from basics, never bluff.